ABDALLAH BOUTRIF

Senior Gameplay / C++ Programmer — Unreal Engine 4 & 5

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PROFESSIONAL SUMMARY

Senior C++ gameplay programmer with seven years in Unreal Engine, five of them in AAA production across PC, Gen 8, and Gen 9 consoles. Shipped and credited on **Dune: Awakening** as Senior Programmer, **Commandos: Origins** as UI Programmer, and **Icarus: Console Edition**. Deep across the engine — multiplayer networking and client prediction, the Gameplay Ability System and custom ability frameworks, Slate/UMG, the reflection system and Blueprint/C++ boundary, and memory and frame-time optimization for console certification.

PROFESSIONAL EXPERIENCE

GRIP Studios — Senior C++ / Gameplay Programmer

Apr 2025 – Present · Hybrid

Co-development and porting studio delivering full console ports and co-developed titles for external clients.

Unannounced AAA Cooperative Shooter (UE5) — for a Sony first-party studio

- Sub-team lead for five programmers: authored technical briefs for system implementations, ran code reviews, and mentored a junior programmer.
- Built a data-object-driven targeting system with designer-authored rules; multi-threaded it as rule counts grew, cutting cost from **15.5 ms to 2.3 ms per frame** — from an entire 60 fps budget down to under 15% of it.
- Rebuilt the gib system from scratch, cutting replication from four properties including visual data structs down to **a single enum and one initial-only target vector**.
- Authored input buffering at the input layer, with native buffering support on Gameplay Abilities.
- Authored the player interaction system and its interactable objects: doors, levers, pickups, and quest items.

Icarus: Console Edition (UE4) — for RocketWerkz · *Shipped — credited*

- Brought in weeks before launch to author a complete, designer-tunable aim assist from scratch — slowdown, stickiness, magnetism, and input countering — computed asynchronously with full debug visualization.
- Stayed through launch resolving console crashes and memory allocation failures, including FMOD leaks that were killing audio on console.

Game Franchise Porting, Gen 8 → Gen 9 (proprietary engine) — for a confidential client studio

- Investigated and documented the engine's PhysX implementation and cloth simulation, assessing usability and per-frame cost on target hardware.
- Owned and documented the cook pipeline, tracking down obscure asserts and asset failures including missing shader compilations.

GamesFarm — Senior Programmer

May 2023 – Apr 2025 · Remote · Promoted from UI Programmer

Commandos: Origins (UE5) — for Claymore Game Studios · *Shipped — credited as UI Programmer*

- Refactored the in-house UI framework end to end and moved it from Blueprint to C++, decoupling UI from gameplay into a stateless MVVM architecture.
- Eliminated hard Blueprint references that were pulling entire character asset chains into memory: character portraits dropped from **2.4 GB to 4 MB** and enemy markers from **3 GB to 42 KB**, reclaiming roughly **5.4 GB** against the Gen 8 memory budget.
- Authored the main menu, settings, tutorial, and collectibles screens, plus in-world 3D interactable UI backed by a widget pool.

Dune: Awakening (UE5) — for Funcom · *Shipped — credited as Senior Programmer*

- Rewrote the Landsraad UI in full and authored a hash-map-driven armor filtering system for minimal CPU cost.

- Unified every button widget into a single class and corrected all usages across the project.
- Authored the death screen, improved character creation, and resolved performance, input, and UI defects ahead of launch.

QORPO — Gameplay Programmer

May 2021 – May 2023 · Hybrid

Two concurrent UE5 productions: a PvP third-person shooter and a creature-collection prototype.

- Architected a shared modular gameplay framework consumed by both projects, so systems were authored once and reused across concurrent productions.
- Authored an in-house ability system from the ground up — ability definition, execution, and replication — along with the ability set built on top of it.
- Implemented gameplay animation systems, vehicles, and interaction systems.
- Authored the technical briefs defining the implementation contract for these systems across the wider team.
- Ran code reviews across a seven-person team and acted as team lead in the lead's absence.

HIGHLIGHTED PROJECT

RAGE — 6DoF Multiplayer Space Combat (UE 5.7) Jun 2026 – Present

Personal project — Creator & Lead Programmer

- Architecting a third-person, six-degrees-of-freedom multiplayer space combat sim from scratch in UE5 C++.
- Networked gameplay via the Gameplay Ability System, with a client-predicted movement system and custom physics sub-stepping for high-velocity flight.
- Built a fully client-predicted projectile class using client-server reconciliation and catch-up simulation to keep high-speed projectiles consistent under latency.

EDUCATION

MSc, Computer Programming and Games Development — Birmingham City University

TECHNICAL SKILLS

Engines & Languages: Unreal Engine 5, Unreal Engine 4, proprietary engines, C++, Blueprints

Unreal Engine: Gameplay Ability System and custom ability frameworks · replication, client prediction & network optimization · Slate / UMG / MVVM ViewModel · Niagara · reflection system (UHT, UPROPERTY/UFUNCTION metadata specifiers) · Blueprint-to-C++ conversion · PhysX / Chaos & cloth · build, cook & platform packaging

Platforms: PC, Gen 8 (PS4, Xbox One), Gen 9 (PS5, Xbox Series X|S)

Profiling & Debugging: Unreal Insights, Unreal profiling tools, console debuggers, memory budgeting for console certification

Core Competencies: Multiplayer networking, gameplay systems architecture, UI authoring & optimization, memory & performance profiling, console porting, technical authoring, code review & mentoring

Tools: Perforce, Git, Diversion, Swarm, Jira, Confluence, Redmine, Slack, VS2017, VS2022, Rider, PowerShell, Steamworks, FMOD, DLSS/XeSS/FSR